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Downscaling of surface air temperature over the Tibetan Plateau based on DEM

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ABSTRACT

Surface air temperature (T_a) is critical to the studies of radiation balance, energy budget, and water cycle. It is a necessary input for associated models. Most of the current T_a datasets of reanalysis products have limitations at local scales due to their coarse spatial resolutions. For better modeling the radiation balance, energy budget, and water cycle over the Tibetan Plateau, this study proposes a practical method for T_a downscaling based on the digital elevation model. This method is applied to downscale T_a of the China regional surface meteorological feature dataset (CRSMFD) at 0.1° and the ERA-interim (ERAI) product at 0.125° to 0.01°. The daily mean T_a and the 3-hourly instantaneous T_a with a 0.01° are obtained. The downscaled T_a are evaluated from the perspectives of accuracy and image quality. Results show that the daily mean T_a downscaled from the CRSMFD product has a RMSE of 1.13 ± 1.0 K at 105 meteorological stations and RMSEs of 0.96 K to 2.34 K at three experimental stations; the instantaneous T_a downscaled from CRSMFD has RMSEs of 1.02 K to 4.0 K at the three experimental stations. T_a after downscaling has better agreement with the ground measured T_a than before downscaling, especially in mountain areas. By contrast, T_a downscaled from the ERAI product has unacceptable accuracy due to the great uncertainty of the ERAI T_a over the Tibetan Plateau. With the proposed method, a 0.01° T_a dataset from 2000 to 2015 over the Tibetan Plateau was generated to satisfy related studies and applications.

1. Introduction

Surface air temperature (T_a) is a key parameter for radiation balance, energy budget, and water cycle studies at regional and global scales. T_a data is an important input for the modeling of land surface processes (Sicart et al., 2008), such as surface evapotranspiration estimation (Kosa, 2009), agricultural fields monitoring (Juknys et al., 2011), and climate change analysis (Jones et al., 1999). The traditional way to obtain T_a is to measure it at ground stations. However, the point-measured T_a cannot reflect the temperature of a large area (Zhou et al., 2017). One possible way to extend T_a measured at spatially distributed ground stations to large areas is spatial interpolation, in which one frequently employed factor is the distance (Dodson and Marks, 1997). In addition to consider the spatial correlation of the T_a variation and the distance, scientific communities have further developed advanced interpolation methods by incorporating more factors, e.g. elevation, latitude, and longitude (Willmott and Matsuura, 1995). With the development of climatic models, long-term T_a products, e.g. National Centers for Environmental Prediction and National Center for Atmospheric Research (NCEP/NCAR) reanalysis dataset and ERA-interim reanalysis dataset (Kalnay et al., 1996; Onogi et al., 2007; Wang et al., 2015), have been generated and are available to users. Compared to the T_a maps interpolated based on ground measurements, these T_a products are usually global or regional and have spatial resolutions ranging from 0.0625° to 1° or even coarser.

Although T_a generated through spatial interpolation and derived from the reanalysis products have succeeded in many applications, they have limitations in some applications. On the one hand, the spatial interpolation is not applicable in areas with insufficient ground stations, e.g. the Tibetan Plateau. On the other hand, T_a provided by the climatic models have exhibited good performance at macro-scales (e.g. continental or global scales), but their coarse resolutions are not suitable for applications at local scales (Hofer et al., 2015). Coarse spatial resolution cannot satisfy the descriptions of spatial variation of T_a at local scales, especially in mountain areas with drastic changes of terrain (Pan et al., 2012).

In order to satisfy the applications in local areas, scientific communities have focused on how to obtain the T_a data with medium to high spatial resolutions. Statistical downscaling has been proven to be an efficient tool to enhance the spatial resolution of T_a. For example,

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Schoof and Pryor (2001) examined the performance of the regression model and neural network model in T_a downscaling; T_a downscaled through these two models were found to yield similar results. By selecting the central and western Europe as the study area, Huth (2002) developed a statistical method to downscale the daily T_a measured by a network of stations, guiding many scientists to start to focus on obtaining T_a with medium spatial resolutions through downscaling. Pan et al. (2012) used the Weather Research and Forecasting (WRF) model to generate a 5 km/1 h T_a dataset to drive the hydrological model in the Heihe River Basin, China. Hofer et al. (2015) used a statistical downscaling method to obtain the daily T_a in a data-sparse glaciated mountain environment. Jha et al. (2015) proposed a geostatistical framework for T_a downscaling. In addition to the statistical downscaling, machine learning methods can also perform well in T_a downscaling (Coulibaly et al., 2005). Additional to the downscaling methods, it has been found that the factors describing T_a variation are also important (Rong et al., 2011). For example, Holden et al., (2011) fully considered the influence of the terrain on the nocturnal T_a in the downscaling of the daily minimum T_a; Kettle and Thompson (2004) used the ground measured T_a and the elevation to downscale the reanalysis T_a data in European mountain areas.

The Tibetan Plateau, which is known as “the roof of the world” and “the third pole” on the Earth, is a focus region of the climate change. T_a is a necessary parameter in the examination of the surface radiation balance and energy budget as well as the water cycle of the Tibetan Plateau. However, the acquisition of T_a over the Tibetan Plateau faces great challenges due to insufficient ground stations and complex terrain. Although there are some T_a datasets provided by reanalysis products, T_a with medium to high resolutions are still rare. Under this context, the objective of this study is to develop a practical method to downscale T_a provided by the current reanalysis products and to reconstruct a long-term daily mean/3-hourly instantaneous T_a dataset with a 0.01°spatial resolution. This T_a dataset can contribute to better modeling the radiation balance, energy budget, and water cycle over the Tibetan Plateau.

2. Study area and datasets

2.1. Study area

The study area is the Tibetan Plateau (73 °E–106 °E, 40 °N–23 °N). As the highest plateau in the world, it has the largest glaciers except the Arctic and Antarctic; thus, it is the birthplace of many rivers in Asia and the source of water for billions of people. The Tibetan Plateau has many mountain areas with steep terrain and varied topography. Its elevation varies from 60 m to over 8000 m, and the average elevation is much higher than the other areas at the same latitude. The digital elevation model (DEM) of the Tibetan Plateau is shown in Fig. 1. Due to complex environment, the Tibetan Plateau has significant impacts on climate change of the surrounding areas and even the whole world. Because of its special location and terrain, the examination of the radiation balance and energy budget and water cycle of the Tibetan Plateau are particularly important. Thus, the scientific communities require the long-term T_a datasets with acceptable accuracy and medium to high spatial resolutions.

2.2. Reanalysis datasets

Two T_a datasets are selected as the basis of downscaling in this study. The first one is the China Regional Surface Meteorological Feature Dataset (CRSMFD) (Chen et al., 2011; Yang et al., 2010), which is based on the existing Princeton reanalysis data, Global Land Surface Data Assimilation System (GLDAS) data, NASA GEWEX Surface Radiation Budget (GEWEX-SRB) radiation data, and Tropical Rainfall Measuring Mission (TRMM) precipitation data. The 6-hourly instantaneous T_a at some meteorological stations operated by China Meteorological Administration (CMA) are also fused in CRSMFD. In addition to T_a, CRSMFD provides atmospheric variables including the near-surface pressure, near-surface air-specific humidity, near-surface wind speed, downward shortwave radiation, downward longwave radiation, and surface precipitation. Here, the T_a dataset is used. CRSMFD has a spatial resolution of 0.1° and a temporal resolution of 3 h. It was downloaded from the Third Pole Environment Database (http://en.tpedatabase.cn/portal/index.jsp).

The second dataset is the global ERA-interim (ERAI) of the European Centre for Medium-Range Weather Forecasts (ECMWF). The original spatial resolution is a reduce Gaussian grid (N128) with an approximately uniform spacing of 79 km (Dee et al., 2011; Gao et al., 2017; Uppala et al., 2008). Using interpolation techniques, the ERAI of original spatial resolution is interpolated to a variety of lon/lat grids from 0.125° to 2.5°. The ERAI used here has a spatial resolution of 0.125° and a temporal resolution of 3 h. It was downloaded from ECMWF (https://www.ecmwf.int/). To satisfy the modeling of surface radiation balance and energy budget of the Tibetan Plateau from 2000 to 2015, the CRSMFD and ERAI datasets during this period were collected.

2.3. Auxiliary datasets

The DEM acquired by the Shuttle Radar Topography Mission was collected to present the terrain of the study area, as shown in Fig. 1. It has a spatial resolution of 90 m. The Normalized Difference Vegetation

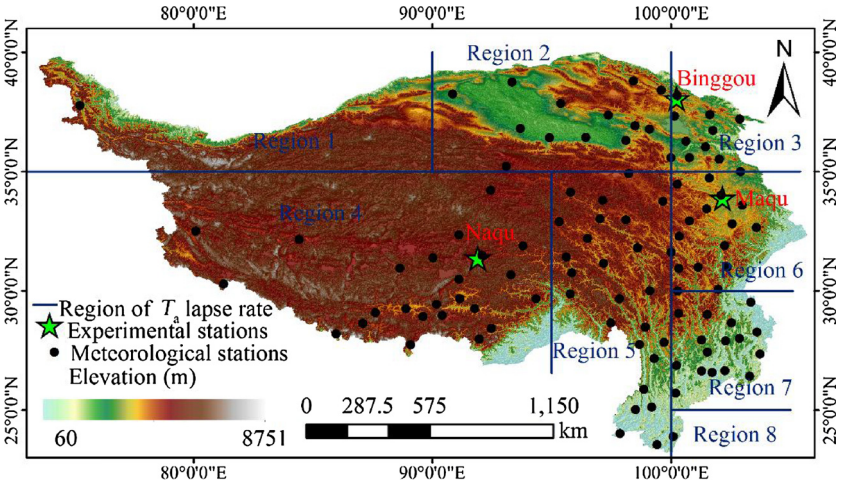


Fig. 1. Digital elevation model (DEM) of the Tibetan Plateau. The meteorological and experimental stations providing the measured T_a are also shown.

Table 1
The sources of the T_a datasets, DEM, and NDVI used in this study.

Data	Source	Website
CRSMFD	Third Pole Environment Database	http://en.tpedatabase.cn/portal/index.jsp
ERAI	European Centre for Medium-Range Weather Forecasts	https://www.ecmwf.int/
DEM	Consortium for Spatial Information (CGIAR-CSI)	http://www.cgiar-csi.org/
NDVI	Land Processes Distributed Active Archive Center (LP DAAC)	https://lpdaac.usgs.gov/

Index (NDVI) provided by the 16-day MODIS vegetation indices product (MOD13A2/MYD13A2) with a 1-km resolution was downloaded from the Land Processes Distributed Active Archive Center (LP DAAC) (https://lpdaac.usgs.gov/). The near-surface air-specific humidity from CRSMFD was also used. Its spatial resolution is consistent with the CRSMFD T_a dataset. The sources of the DEM, NDVI, and the aforementioned CRSMFD and EARI are listed in Table 1.

In order to evaluate the downscaled T_a, two categories of ground measured T_a were collected. The first category is the daily mean T_a at 107 meteorological stations of CMA in the study area in 2010 released by the China Meteorological Data Service Center (http://data.cma.cn/en). Locations of these stations are shown in Fig. 1. Note that the instantaneous T_a at these stations were not available. The second category is the instantaneous T_a measured at three experimental stations, including Maqu, Naqu, and Binggou. Details of these three sites are shown in Table 2. Maqu and Naqu stations are located in relatively flat areas; Binggou station is located in a valley with mountains to its east and west. Since the 107 CMA stations and the three experimental stations yield typical terrain in the study area, their measurements would be beneficial for the evaluation of the downscaled T_a.

3. Methodology

3.1. Influencing factors of T_a

The hypothesis of this study is that the main factors affecting the daily mean T_a and the instantaneous T_a at Tibetan Plateau are surface characteristics and the location, e.g. the elevation and latitude. Based on the daily mean T_a of the aforementioned 107 meteorological stations and the instantaneous T_a derived from the CRSMFD dataset, this hypothesis is tested through the following approach. First, linear regression is conducted with the daily mean T_a /instantaneous T_a as the dependent variable and the possible factors as the independent variables; different combinations of the possible factors are tested, including (1) Case I: elevation, (2) Case II: elevation and latitude; (3) Case III: elevation, latitude, and longitude; and (4) Case IV: elevation, latitude, longitude, and NDVI. Elevation is selected as the main factor because T_a decreases with increasing elevation, especially in mountain areas.

Table 2
The three experimental stations providing the T_a measurements.

Station	Interval of observation (min)	Elevation (m)	Source
Maqu	30	3435	NIEER[a]
Naqu	10	4512	CEOP-AEGIS[b]
Binggou	10	3449	Hi WATER[c]

[a] Northwest Institute of Eco-Environment and Resources Chinese Academy of Sciences (Shang et al., 2016).

[b] Coordinated Asia-European long-term Observing system of Qinghai-Tibet Plateau Hydro-meteorological processes and the Asian-monsoon systEm with Ground satellite Image data and numerical Simulations (Wang et al., 2012).

[c] Heihe Watershed Allied Telemetry Experimental Research (Li et al., 2013a, 2009).

Latitude and longitude represent the location and the solar angle. NDVI is used here to parameterize the vegetation coverage. T_a is also affected by other surface characteristics such as the soil moisture and precipitation (Schwingshackl et al., 2017; Yao et al., 1999). However, the soil moisture and precipitation are not taken into account in this study, because the current products have coarse spatial resolutions and temporal resolutions (Gebremichael et al., 2008; Piles et al., 2011) and cannot satisfy T_a downscaling. Regression for the daily mean T_a is conducted for that provided by the ground stations, among which two stations are excluded due to abnormal T_a values and the rest 105 stations are retained. Since the instantaneous T_a at these stations are not available, regression for the instantaneous T_a is conducted for that derived from the CRSMFD dataset. The coefficients of determination (R^2) in the linear regression are shown in Fig. 2.

Significant correlations ($p < 0.01$) are found between T_a and the possible factors. As for the daily mean T_a, Fig. 2(a) demonstrates that the elevation in Case I explains 50%–76% of the spatial variation of T_a over the entire study area. Thus, the elevation can play as the main factor in T_a downscaling. We also find that the latitude also affects the spatial variation of T_a, as reported by Li et al. (2013b). 75%–93% of T_a can be explained by the elevation and latitude. By contrast, for the 105 stations, the contributions of the longitude and NDVI to the spatial variation of T_a are ignorable: the maximum increments of R^2 are 0.03 and 0.01 in Case III and Case IV, respectively.

Similar results are found for the instantaneous T_a. The result at 06:00 UTC are shown in Fig. 2(b) as an example. In Case I, the elevation explains 51%–84% of the spatial variation of T_a. In Case II, R^2 increases to 0.78-0.86. In Case III, adding the longitude as the independent variable induces a 0.01-0.02 increment of R^2. However, in Case IV, adding NDVI only contributes to a maximum increment of R^2 of 0.01. From the aforementioned regression, influences of the elevation, latitude, and longitude on the T_a can be quantified. For example, the influence of the elevation is approximately 4–7 K/km; the influences of the latitude and longitude are approximately 0.6–1.5 K/latitude and 0.3–1.0 K/longitude, respectively.

3.2. Downscaling of T_a

According to the previously described linear relationship between T_a and its influencing factors, T_a can be expressed as:

$$T_{a,daily} = f_{daily}(H, X_1) = \lambda H + aX_1 + c \quad (1)$$

$$T_{a,ins} = f_{ins}(H, X_1, X_2) = \lambda H + aX_1 + bX_2 + c \quad (2)$$

where $T_{a,daily}$ and $T_{a,ins}$ are the daily mean and instantaneous T_a in K, respectively; f_{daily} and f_{ins} are the statistical functions for the daily mean T_a and instantaneous T_a, respectively; H, X_1, X_2 are the elevation, latitude, and longitude, respectively; λ, a, and b are the corresponding coefficients; and c is the intercept. It is evident that λ is the lapse rate (LR) of T_a (Du et al., 2010; Fang and Yoda, 1988). Note that the longitude is not contained in Eq. (1) due to its ignorable ability in explaining the daily mean T_a as analyzed in Section 3.1.

Based on Eqs. (1) and (2), the flowchart of the proposed method for T_a downscaling is shown in Fig. 3. The first stage for T_a downscaling is to calculate LR. Coefficient a, b, and c will be analyzed later. The DEM data at 90-m is aggregated to 0.01°. The mean elevation of the 10 × 10 pixels is calculated and used as the elevation of the pixel at 0.1° that containing these 10 × 10 pixels. According to examinations on T_a from 1962 to 2011 by Li et al. (2013b), the spatial distribution of LR can be divided into eight regions, i.e. Region 1: 73–90 °E, 35–40 °N; Region 2: 90–100 °E, 35–40 °N; Region 3: 100–105 °E, 35–40 °N; Region 4: 78–95 °E, 27–35 °N; Region 5: 95–100 °E, 27–35 °N; Region 6: 100–107.5 °E, 30–35 °N; Region 7: 100–105 °E, 25–30 °N; Region 8: 100–105 °E, 23–25 °N. These regions are shown in Fig. 1. In this division scheme, each region has similar regional climatic characteristics and a range of elevation changes. This division scheme is utilized by

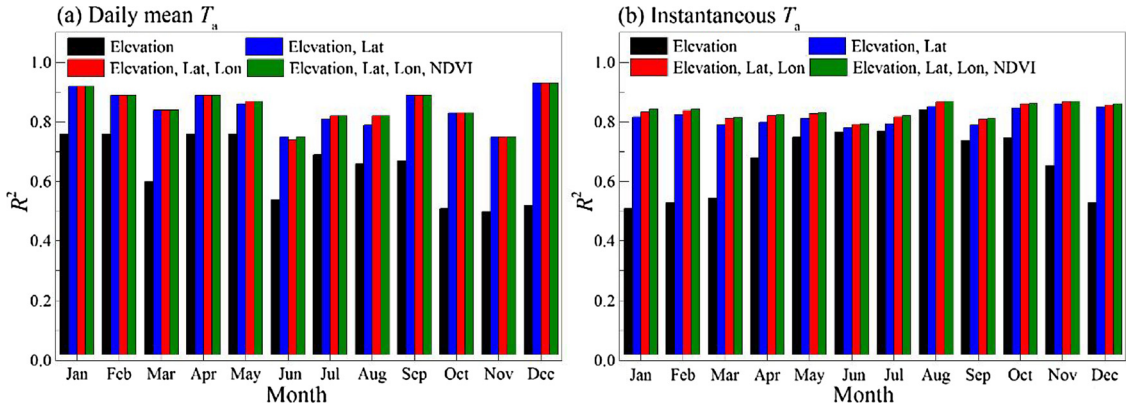


Fig. 2. Coefficient of determination (R^2) in the linear regression between T_a and different factors in the study area.

this study. To better address the intra-annual variations of LR, the LR values of the instantaneous T_a at every 3 h and the daily mean T_a on every day are calculated.

The second stage is to determine and optimize the initial value of T_a at the target resolution. The T_a value at the native resolution (i.e. 0.1°) is taken as the initial value of the pixel at the target resolution (i.e. 0.01°). At the target resolution, a moving window approach is employed to refine the initial T_a of the central pixel. For each pixel at the target resolution, the window size is set to $n \times n$ pixels (where n is an odd number) and the current pixel under consideration is the center of the window. If the current pixel is on the edge of the image, the window is not complete and the existing pixels are selected. Pixels with valid T_a and elevation in the moving window are selected as valid pixels. Then the mean T_a of the valid pixels in the moving window is calculated as the optimized T_a of the central pixel as follows:

$$T'_a = \frac{\sum_{i=1}^{m} T_{a\text{-initial}}(i)}{m} \quad (3)$$

where T_a' is optimized initial value of the T_a; $T_{a\text{-initial}}$ (i) is the initial T_a the i-th pixel at the target resolution within the window; and m is the number of valid pixels in the window.

The third stage is to determine the final value of T_a at the target resolution. According Eqs. (1) and (2), the T_a difference between the central pixel of moving window and the mean T_a of moving window can be expressed as:

$$\Delta T_{a,daily} = \lambda (H - H_{win}) + a (X_{1-i} - X_{1-win}) \quad (4)$$

$$\Delta T_{a,ins} = \lambda (H - H_{win}) + a (X_{1-i} - X_{1-win}) + b (X_{2-i} - X_{2-win}) \quad (5)$$

where $\Delta T_{a,daily}$ and $\Delta T_{a,ins}$ are daily mean T_a difference and instantaneous T_a difference in K; H, $X_{1\text{-}i}$ and $X_{2\text{-}i}$ are the elevation, latitude, and longitude of the central pixel of the moving window, respectively; H_{win}, $X_{1\text{-win}}$ and $X_{2\text{-win}}$ are the mean elevation, latitude, and longitude of the moving window, respectively.

$X_{1\text{-}i}$ and $X_{1\text{-win}}$, $X_{2\text{-}i}$, and $X_{2\text{-win}}$ can be considered to be approximately equal (Duan et al., 2017). Thus, Eqs. (4) and (5) can be simplified as:

$$\Delta T = \lambda (H - H_{win}) \quad (6)$$

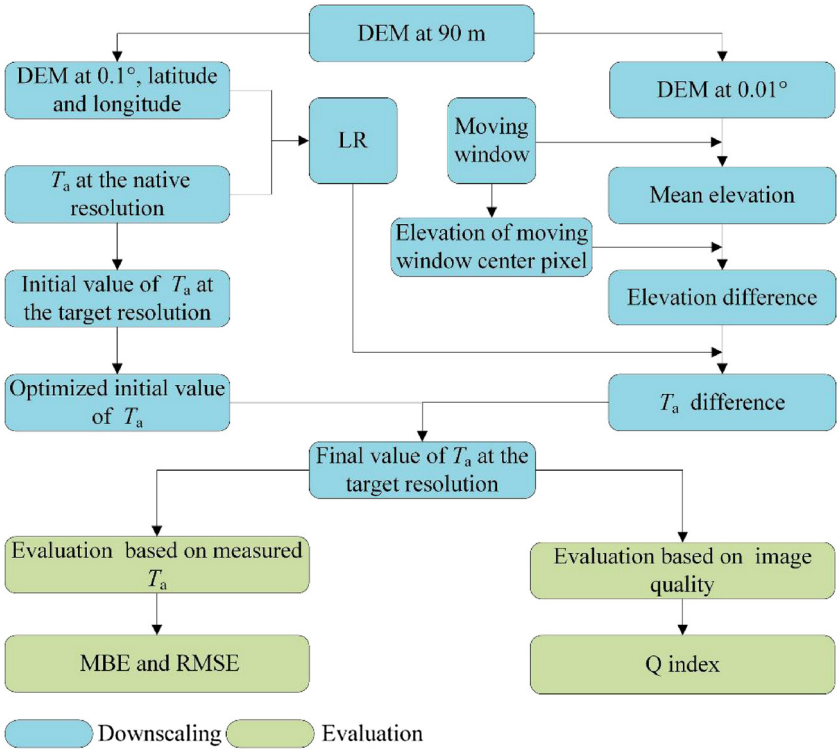


Fig. 3. Flowchart of the proposed method for T_a downscaling.

where ΔT is the T_a difference in K.

Then the final T_a of the central pixel is:

$$T_a = T'_a + \Delta T \quad (7)$$

3.3. Evaluation of the downscaled T_a

The proposed downscaling method is applied to the daily mean T_a and instantaneous T_a derived from the CRSMFD and ERAI datasets. Note that the instantaneous T_a provided by these two datasets is 3-hourly. Thus, the daily mean T_a is calculated through directly averaging the instantaneous T_a. Both the daily mean T_a and the instantaneous T_a are downscaled to 0.01°. On the one hand, the downscaled daily mean T_a is validated against the daily mean T_a of the 105 meteorological stations and the three experimental stations, including Maqu, Naqu, and Binggou. On the other hand, the downscaled instantaneous T_a is validated based on the measurements at the three experimental stations. The mean bias error (MBE) and root mean squared error (RMSE) are used as validation indexes. MBE can reflect the overestimation or underestimation, while RMSE can measure the deviation between the estimate and the true value. MBE and RMSE are calculated as follows:

$$MBE = \frac{\sum_{i=1}^{n}(T_a - T_{a,\text{in-situ}})}{n} \quad (8)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(T_a - T_{a,\text{in-situ}})^2}{n}} \quad (9)$$

where $T_{a,\ \text{in-situ}}$ is T_a measured at the ground stations; and n is the number of samples.

In addition to the accuracy, the downscaled T_a is further evaluated from the perspective of image quality. The image quality index (Q) is calculated to evaluate the image quality of the downscaled T_a. Q index, which is a powerful parameters to evaluate the loss of correlation, luminance distortion, and contrast distortion, is frequently employed in the evaluation of the downscaled land surface temperature (Wang and Bovik, 2002; Zhou et al., 2016). Q is calculated as follows:

$$Q = \frac{4\delta_{OD}O_T D_T}{(\delta_O^2 + \delta_D^2)[(O_T)^2 + (D_T)^2]} \quad (10)$$

where O_T and D_T are the mean T_a values before and after downscaling, respectively; δ_o^2 and δ_D^2 are their variances; δ_{OD} is their covariance between the T_a images before and after downscaling. A moving window method with sizes of 5 × 5 pixels, 10 × 10 pixels, 15 × 15 pixels, 20 × 20 pixels, 50 × 50 pixels, 100 × 100 pixels, 150 × 150 pixels is used to calculate Q.

4. Results and discussion

4.1. LR and the window size in downscaling

LR, which is a necessary parameter in the proposed T_a downscaling method, is calculated based on the CRSMFD data. The annual mean LR values of the eight regions for 2010 are shown in Table 3 as examples.

Table 3
The calculated annual mean LR values (unit: K/km), standard deviation (STD), annual mean air specific humidity (g/kg), and mean elevation (m) for the eight regions in 2010.

Parameter	Region							
	1	2	3	4	5	6	7	8
LR	5.78	6.74	5.21	5.15	4.90	4.35	4.06	4.82
STD	1.02	0.85	1.31	0.49	0.68	0.80	0.91	1.72
Specific humidity	3.786	3.050	4.190	3.328	4.968	5.740	8.361	9.970
Elevation	4498	3768	3257	4825	3893	3547	2622	1794

For the study area, LR ranges from 4.35 ± 0.80 K/km (Region 6) to 6.74 ± 0.85 K/km (Region 2). The calculated LR values are compared with those reported by Li et al. (2013b). We find that these two LR results agree well with each other in tendency. Nevertheless, slight differences in magnitude are found. One possible reason is that different T_a datasets are used in our study and Li et al. (2013b), which calculated the LR values based on T_a measured at the meteorological stations. According to Fig. 1, one can find that there are a few meteorological stations in some regions, e.g. Regions 1, 4, and 8, for which the calculated LR may have great uncertainty.

Differences among the LR values of different regions stimulate us to conduct further analysis. As shown in Table 3, the tendency of the LR has not evident correlation with the elevation. However, the LR shows a contrary distribution with the air specific humidity. Regions with higher LR always have drier air, consistent with previous studies (Blandford et al., 2008; Minder et al., 2010; Tang and Fang, 2006). Among the eight regions, Regions 1, 2, 3, and 4 yield higher LR values than the other four regions. One main reason for this phenomenon is the lower air humidity in these four regions.

A closer look on Table 3 demonstrates a standard deviation (STD) 0.49–1.72 of the LR for the eight regions. Higher STD values are observed for Region 3 (1.31) and Region 8 (1.72), indicating high intra-annual variation of LR in T_a downscaling. This finding is further confirmed by Fig. 4. The obvious seasonal behavior of LR can be seen from Fig. 4. The LR in the monsoon season of regions 1, 2, 3, 6, and 7 is higher than in winter. LR in other regions also has significant differences in various months. Therefore, it is necessary to use the LR with a finer temporal resolution in T_a downscaling. Thus, in the downscaling of the daily mean T_a, the corresponding LR on the same day is used; in the downscaling of the instantaneous T_a, the mean LR at the corresponding time smoothed with a 5-day window is used to avoid the LR estimation error caused by the data anomaly.

Another important parameter in T_a downscaling is the size of the moving window. In order to find the optimal window size in the downscaling process, 12 sizes ranging from 5 × 5 pixels to 51 × 51 pixels with varying increments and the downscaled T_a at each size is validated at the three experimental stations. RMSE is selected as the index to determine the optimal window size. For both the CRSMFD and ERAI T_a data, the RMSE exhibits a decreasing-increasing varying trend, though no significant changes on the RMSE value are found. An example for Naqu station is shown in Fig. 5. The optimal window size is around 11 × 11 pixels. Note that the native resolution before downscaling is 0.1° and the target resolution after downscaling is 0.01°. Therefore, the ratio between the native resolution and the target resolution is 10. On the one hand, when the window size is smaller than 11 × 11 pixels, the intrinsic downscaling is the separate the coarser pixel (i.e. 0.1°) to 10 × 10 finer pixels (i.e. 0.01°) and the relationship between two neighbor coarse pixels is not considered in the downscaling. Thus, evident boxy effect occurs in the Ta images after downscaling (Zhou et al., 2016), especially in areas with rugged terrain and high elevation fluctuations, such as in the southeastern part of the Tibetan Plateau. Therefore, 11 × 11 pixels is finally used as the size of the moving window in the downscaling of both CRSMFD and ERAI T_a.

4.2. Evaluation of the downscaled daily mean T_a

T_a at 0.01° obtained through downscaling the CRSMFD and ERAI daily mean T_a is firstly validated based on the ground measured daily mean T_a at 105 meteorological stations, after excluding two stations with abnormal recordings of T_a. Because the 6-hourly instantaneous T_a at only part of these stations was fused to generate the CRSMFD T_a dataset (Chen et al., 2011; Yang et al., 2010), the validation of the daily mean T_a is still valid. The scatter plots between the CRSMFD daily mean T_a before and after downscaling and the ground measured daily mean T_a at the 105 meteorological stations on six days in 2010 are shown in Fig. 6. It is evident that the downscaled T_a is closer to the ground

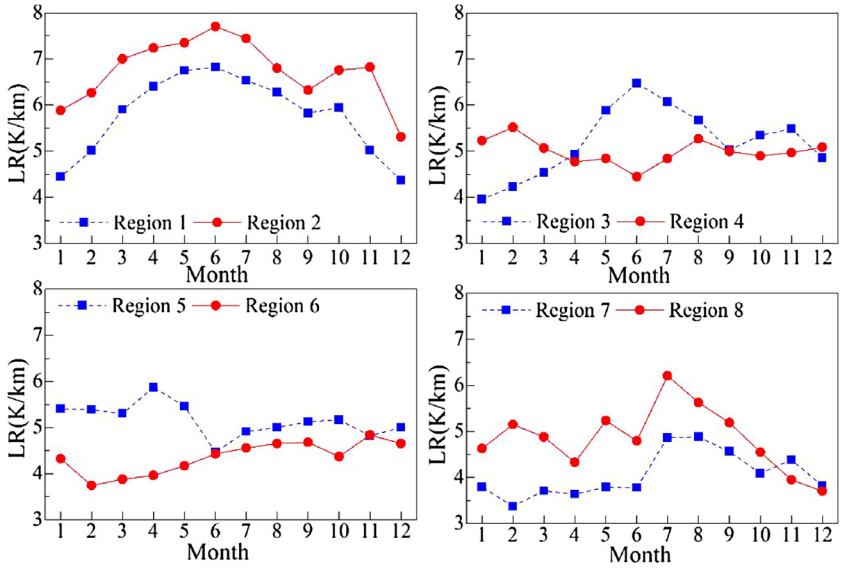


Fig. 4. The monthly mean LR values for the eight regions in 2010.

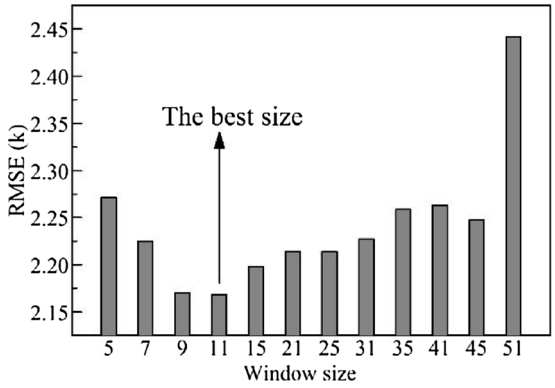


Fig. 5. The RMSE values of the downscaled 3-hourly CRSMFD T_a with different window sizes at Naqu station.

measured T_a than the CRSMDF T_a before downscaling. As for the CRSMFD T_a before downscaling, the MBE values ranges from −1.12 K to −0.81 K, indicating a negative deviation of the CRSMFD daily mean T_a. The corresponding RMSE values ranges from 2.01 K to 2.29 K. By contrast, the MBE and RMSE values of the downscaled CRSMFD T_a range from −0.91 K to −0.52 K and 1.22 K to 1.60 K. Because no calibration of the original CRSMFD T_a was conducted and the downscaled T_a at 0.01° was generated based on the original T_a, the slight negative deviation of the downscaled T_a results from the negative deviation of the T_a before downscaling.

The ground measured daily mean T_a at Maqu, Naqu, and Binggou are further used to validate the downscaled daily mean T_a obtained from CRSMFD. R^2, MBE, and RMSE of these three stations are shown in Table 4. After downscaling, the MBE values at Maqu and Naqu decrease from 2.10 K to 2.0 K and from 0.68 K to 0.44 K; negative deviation at Binggou (MBE) are depressed: the MBE decreases from −1.78 K to 0.05 K. At all these three stations, better agreement between the CRSMFD T_a and the ground measured T_a can be found after downscaling: the RMSE values decreases 0.07 K for Maqu, 0.14 K for Naqu, and 0.78 K for Binggou. The best performance of the downscaling is observed for Binggou station because it is located in a mountain area. Nevertheless, better agreement between the downscaled T_a and the ground measured T_a than T_a before downscaling is mainly contributed from the DEM data with a high spatial resolution. It cannot be concluded that the downscaling can improve the accuracy of the original CRSMFD T_a.

Scatter plots between the ERAI daily mean T_a before and after downscaling and the ground measured T_a are shown in Fig. 7. It is clear that the daily mean T_a calculated based on ERAI product has a significant underestimation over the entire Tibetan Plateau. Before downscaling, the MBE value ranges from −13.87 K to −10.66 K. After downscaling, the significant underestimation still exists (MBE: from −13.05 K to −9.72 K), although the systematic error has been mitigated due to the scale mismatch between the ground station and the pixel decreases.

In order to further analyze the spatial distribution of errors of the daily mean T_a, the annual RMSE values of the daily mean T_a of CRSMFD and ERAI products at the 105 stations are calculated and shown in Fig. 8. For CRSMFD daily mean T_a before downscaling, one can see that larger RMSE values (i.e. ≥ 5.0 K) occurs in the southern and southeastern parts of the Tibetan Plateau while lower RMSE values (i.e. ≤ 1.0 K) in the northern and central parts (Fig. 8a). According to Fig. 1, the terrain in the southern and southeastern parts of the Tibetan Plateau is rugged while the terrain in the other part is relatively flat. In areas with rugged terrain, T_a varies drastically with the elevation; thus, T_a measured at ground stations cannot reflect the true air temperature in the areas at the pixel scale before downscaling (i.e. 0.1° × 0.1°). This finding is further confirmed by Fig. 9, which shows the elevation difference between the station and its surrounding area (window size: 11 × 11 pixels).

Compared with the daily mean T_a before downscaling, the deviation of the downscaled T_a and the ground measured T_a was found to significantly decrease at the stations located in the southern and southeastern parts of the Tibetan Plateau (Fig. 8c). The main reason of such a decrease is the better spatial resolution (i.e. 0.01°) after downscaling. However, relatively higher RMSE values (i.e. 2.0–3.0 K) are still found at the stations with rugged terrain. This phenomenon demonstrates that higher spatial resolution is still needed for local applications in these areas.

In order to exhibit the performance of the proposed downscaling method, T_a before and after downscaling are further compared with T_a resampled from the native resolution to 0.01°. In the resampling, the bilinear interpolation approach is employed and the deviation of resampling T_a is shown in Fig. 8(e) for CRSMFD. Comparison between Fig. 8(a), (c), and (e) clearly shows that the resampled T_a has higher deviation from the ground measured daily T_a than the downscaled T_a. The annual mean RMSE values of these three T_a datasets are 1.40 ± 1.43 K, 1.03 ± 1.0 K, and 1.72 ± 1.66 K. Therefore, it is

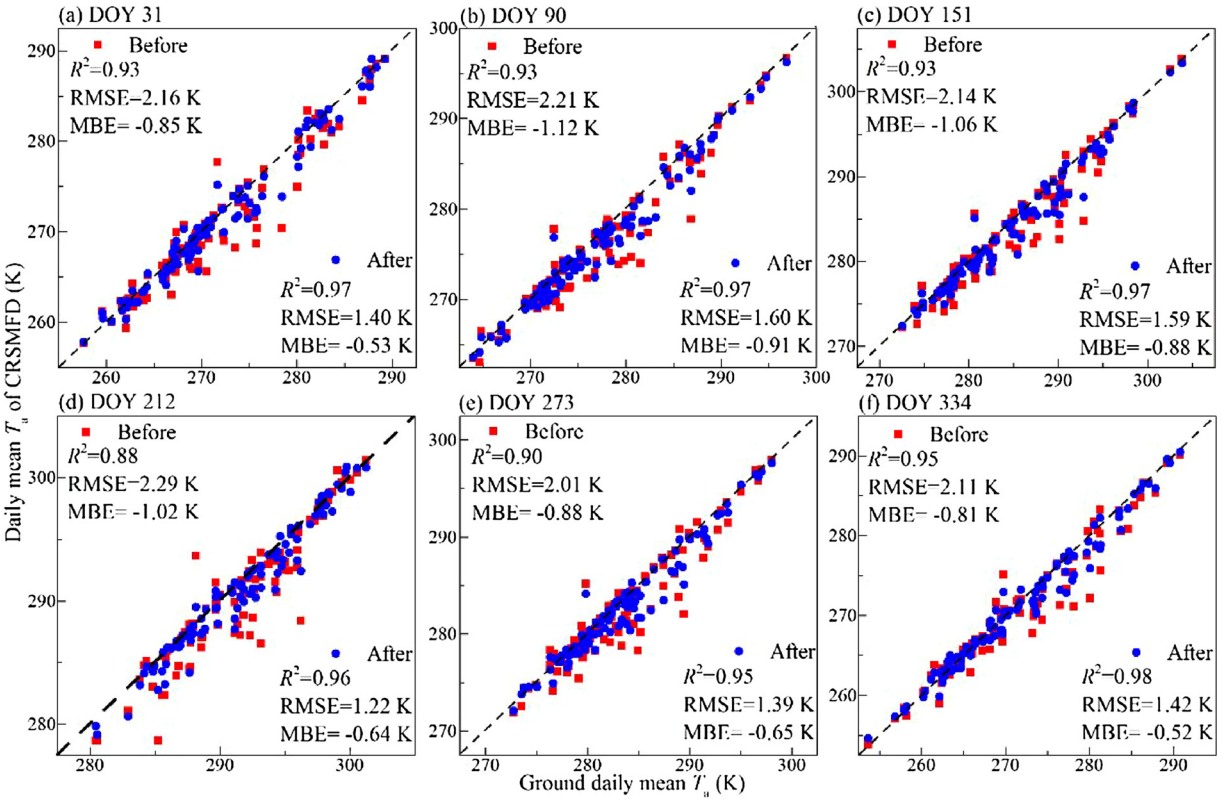


Fig. 6. Validation of the CRSMFD daily mean T_a before and after downscaling based on the ground measured daily mean T_a at the 105 meteorological stations in 2010.

Table 4
Validation of the CRSMFD daily mean T_a before and after downscaling based on the three experimental stations.

Station	Before downscaling			After downscaling		
	R^2	MBE	RMSE	R^2	MBE	RMSE
Maqu	0.98	2.10	2.41	0.99	2.0	2.34
Naqu	0.99	0.68	1.10	0.99	0.44	0.96
Binggou	0.94	−1.78	2.79	0.95	0.05	2.01

reasonable to conclude that the proposed downscaling method has good performance through incorporating DEM data with high spatial resolution.

Due to significant negative deviation of the daily mean T_a derived from ERAI, the dependence of the deviation on the terrain is not evident, as shown by Fig. 8(b). RMSE values of T_a before downscaling, after downscaling, and after resampling are 14.34 ± 15.81 K, 13.32 ± 15.83 K, and 14.34 ± 15.82 K. Only slight decrease of the RMSE is observed for the T_a after downscaling. Therefore, the accuracy of the T_a after downscaling mainly depends on the T_a before downscaling.

4.3. Evaluation of the downscaled instantaneous T_a

In addition to the daily mean T_a, the 3-hourly instantaneous T_a derived from CRSMFD and ERAI were downscaled to 0.01° and then validated against the ground measured T_a at Maqu, Naqu, and Binggou. These three experimental stations are located in areas with different terrain. The differences between the elevations of the stations and the mean elevations of the moving windows are 0.4 m for Maqu, 31 m for Naqu, and −233 m for Binggou, respectively. Results for the CRSMFD T_a are shown in Table 5.

Table 5 demonstrates that the annual mean R^2 values for the 3-hourly instantaneous T_a before and after downscaling at most times exhibit insignificant changes at these three stations, except slight increments at some times. Thus, the CRSMFD T_a before and after downscaling yields good agreements in the intra-annual variations to the ground measured T_a. Despite the good correlation in varying trend, the MBE and RMSE values before and after downscaling exhibit diverse variations. At Maqu station, which is located in a flat area, the CRSMFD T_a yields significant overestimation at nighttime and in the morning (i.e. at 0:00, 03:00, 12:00-21:00 UTC) while slight underestimation in the afternoon (i.e. 06:00 and 09:00 UTC) according to the MBE values. Thus, the T_a after downscaling yields similar systematic error. Although the area is relatively flat, the downscaled T_a yields lower deviation from the ground measured T_a than the T_a before downscaling, indicating that the scale mismatch between the station and the pixel scale is decreased.

At Naqu station, the CRSMFD T_a before downscaling is higher than the ground measured T_a at the time with low temperature (i.e. in the early morning and at nighttime, with MBE values of 1.46 K at 0:00, 0.86 K at 12:00, 1.90 K at 15:00, and 1.85 K at 18:00, and 1.84 K at 21:00) and lower during the daytime (with MBE values of −0.68 K at 03:00, −1.0 K at 06:00, and −0.64 K at 09:00). Because the elevation of Naqu station is relatively higher than the surrounding area, its T_a is slightly decreased compared with the T_a of the moving window when downscaling. Thus, it is understandable that the downscaled T_a yields better agreement with the ground measured T_a when the T_a before downscaling yields positive deviation from the ground T_a, and vice versa. Considering all the eight times, the downscaled T_a yields better agreement with the ground T_a than the original T_a.

Table 5 indicates that the deviation of the CRSMFD T_a before downscaling from the ground measured T_a at Binggou station are different from that at Maqu and Naqu. The CRSMFD T_a is much lower than the ground measured T_a during the entire diurnal cycle except at 09:00. The MBE values ranges from −3.65 K at 0:00 to −0.23 K at 06:00. Accordingly, the RMSE values range from 4.48 K to 2.15 K. As

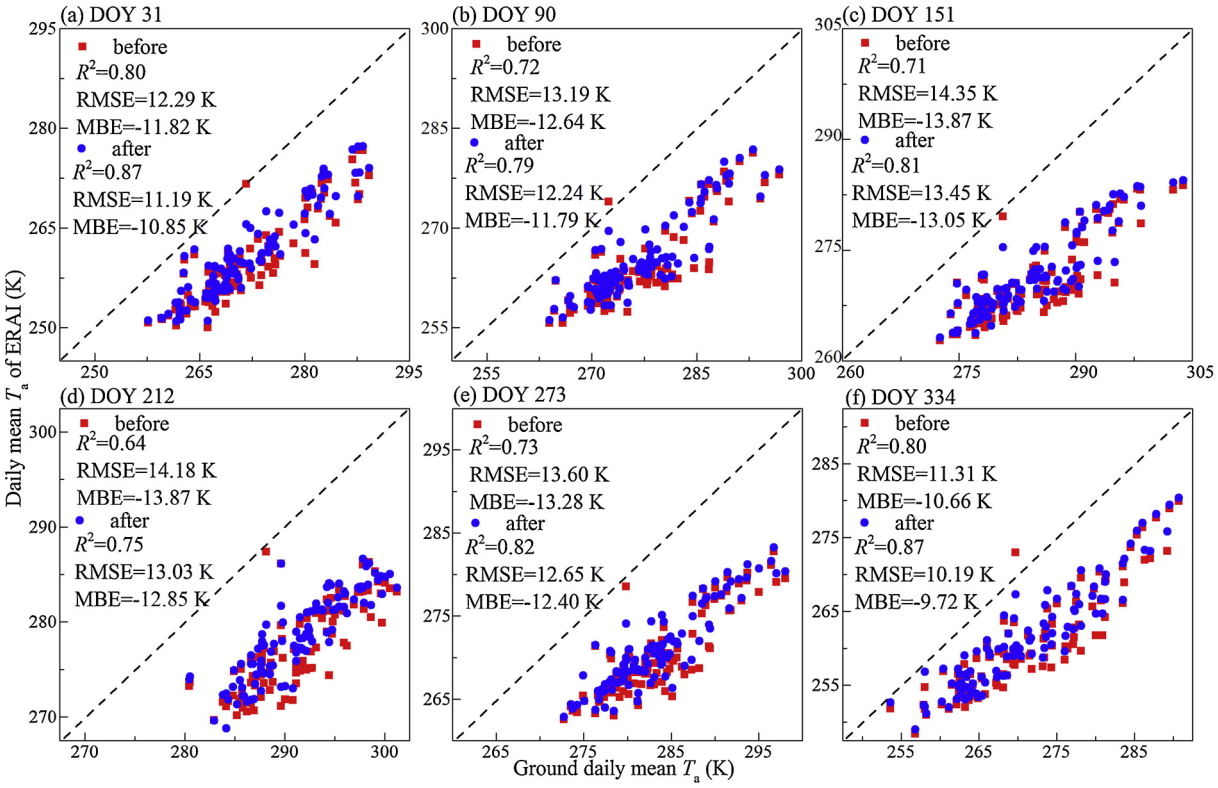


Fig. 7. Validation of the ERAI daily mean T_a before and after downscaling based on the ground measured daily mean T_a at the 105 meteorological stations in 2010.

mentioned previously, the large deviation is mainly induced by the rugged terrain surrounding Binggou station and it exaggerates the scale mismatch between the station and the corresponding pixel with a 0.1° resolution. Due to a lower elevation of Binggou station than its surrounding area, a positive increment of T_a was added to the average T_a of the moving window. Therefore, the downscaled T_a yields better agreement with the ground measured T_a when the original CRSMFD yields largely negative deviation (i.e. at 0:00, 03:00, 06:00, 12:00, 15:00, 18:00, and 21:00).

The ERAI T_a before and after downscaling are also compared against the ground measured T_a at the three experimental stations. Due to space limitation, the results are not listed here. The R^2 value ranges from 0.88 to 0.91 at Maqu site and 0.84 to 0.90 at Naqu for the T_a before and after downscaling, indicating that both of these two T_a have very similar varying trends as the ground measured T_a. However, R^2 at Binggou site decreases to 0.15-0.36 at Binggou site, suggesting that the varying trend of the ERAI T_a is largely different from the ground measured T_a. A closer look at the MBE values of the ERAI T_a indicates that the ERAI T_a before and after downscaling are significantly lower than the ground measured T_a, with MBE values lower than -6.0 K and RMSE values higher than 7.85 K at all the three stations. Considering the performance of the daily mean T_a of ERAI, it should be concluded that users should take the uncertainty of ERAI T_a in the Tibetan Plateau into account in their applications.

4.4. Evaluation of the image quality of the downscaled T_a image

Previous evaluation shows that the ERAI T_a has unacceptable accuracy in the Tibetan Plateau. Therefore, in this section, only the CRSMFD T_a after downscaling is evaluated from the perspective of image quality. The CRSMFD T_a before and after downscaling on four days (i.e. DOY 1, 91, 182, 274) in 2010 are shown in Fig. 10 as examples. Additional, the T_a of two sub-regions, including region A and region B are shown in Fig. 11.

Fig. 10 indicate that the T_a images before and after downscaling are very similar at the plateau scale, including the magnitudes and the spatial patterns: the southeastern and northern parts are warmer due to lower elevations while the western and central parts are cooler due to higher elevations (see Fig. 1 for elevation). A closer look on Fig. 11 indicates that the downscaled T_a presents more details about T_a, especially in the southern and southeastern parts, where there are many mountains. Fig. 11 clears shows more much details about T_a presented by the downscaled T_a. There are mountains and valleys in both regions A and B (Fig. 11). Although the T_a at 0.1° can present the spatial pattern of the T_a in the entire regions, the coarse resolution cannot satisfy the requirement for mapping the T_a at local scales. The downscaled T_a at 0.01° can clear show the spatial distribution of T_a according to the mountains and valleys.

In order to further quantitative evaluate the image quality of the downscaled T_a, the CRSMFD at 0.1° was resampled to 0.01° through the nearest neighbor approach and then Q was calculated between the downscaled T_a and resampled T_a of the entire Tibetan Plateau. Note that the resampled T_a still maintains a high degree of spatial consistency and numerical consistency with the original T_a (i.e. at 0.1°). The Q values were calculated based on moving windows with seven different sizes (5 × 5 pixels, 10 × 10 pixels, 15 × 15 pixels, 20 × 20 pixels, 50 × 50 pixels, 100 × 100 pixels, and 150 × 150 pixels) on six days are shown in Fig. 12. The mean values of the entire Tibetan Plateau are used for comparison. When the window size is 5 × 5 pixels, Q ranges from 0.43 to 0.45, depending on specific day. Such low Q values results from the different T_a details in the windows of the T_a images generated through downscaling and resampling. It is understandable because much more details about the T_a have been added through downscaling while no more information has been added in the resampling. The Q value increases according to the increase of the window size. The Q value is higher than 0.88 with a size of 50 × 50 pixels and 0.92 with a size of 100 × 100 pixels, indicating that shows that the CRSMFD T_a after downscaling maintains a high degree of consistency with the original data in the both the overall spatial distribution and magnitude. Therefore, the downscaling method used in this study can

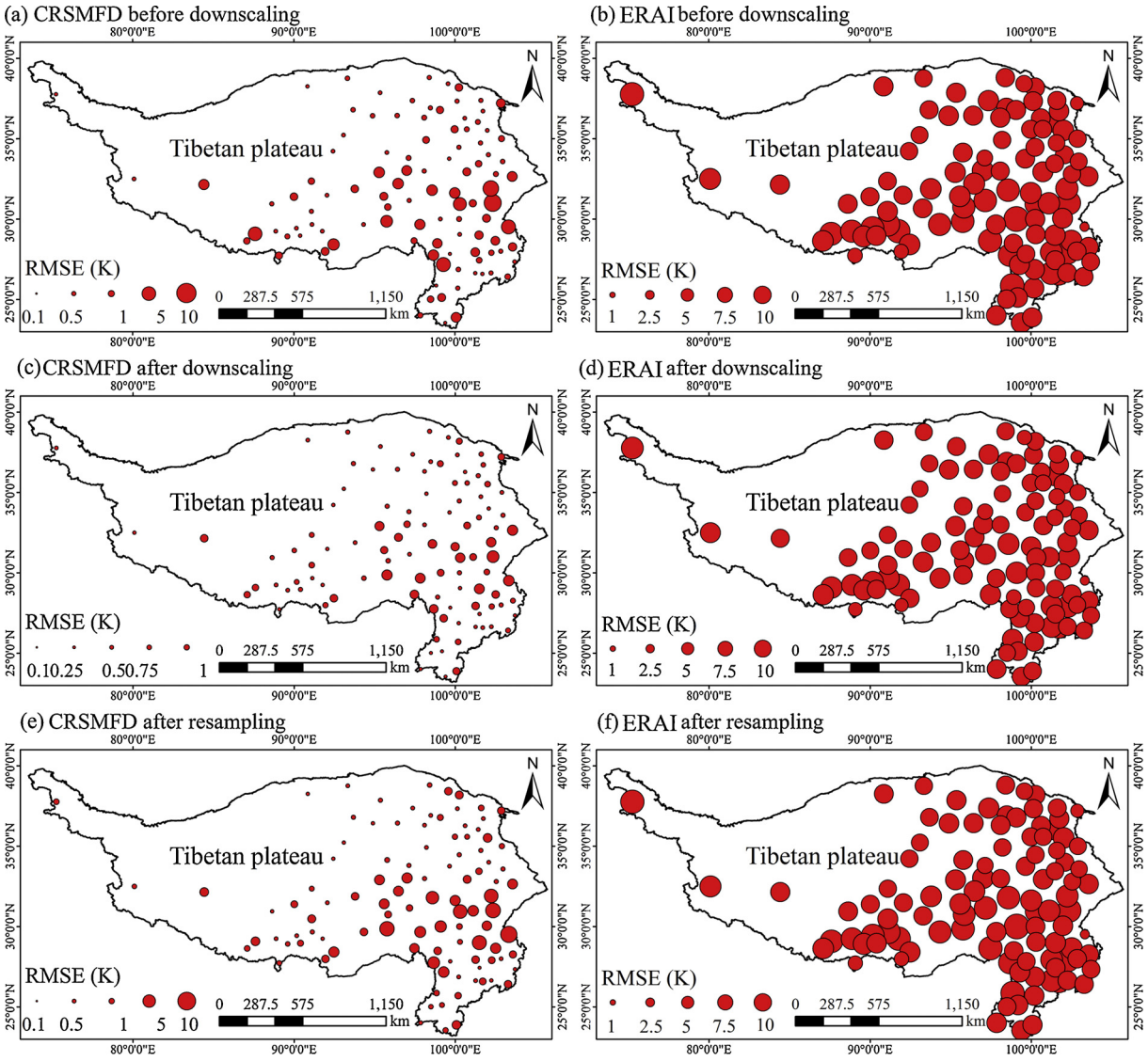


Fig. 8. The annual mean RMSE of CRSMFD and ERAI daily mean T_a at 105 ground sites.

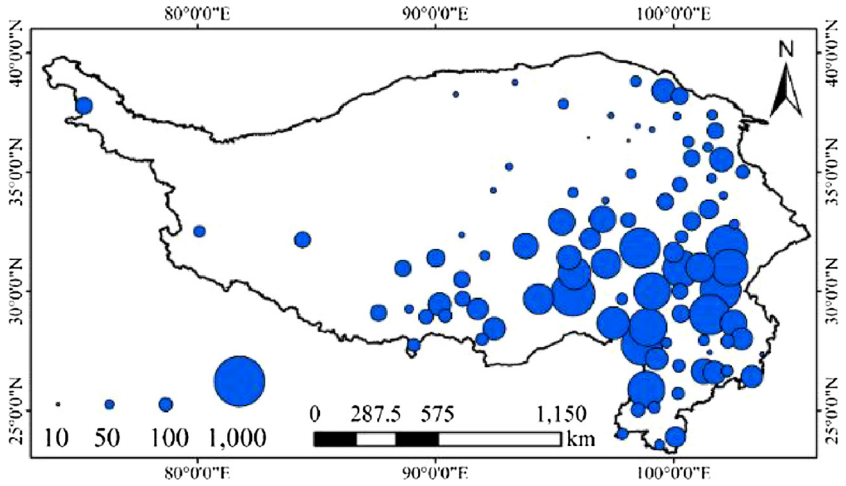


Fig. 9. The absolute elevation difference between the 105 meteorological stations and the surrounding windows.

Table 5
The annual mean R^2, MBE (K), and RMSE (K) of the CRSMFD 3-hourly instantaneous T_a when being validated against the ground measured T_a at Maqu, Naqu, and Binggou.

Time (UTC)	T_a	Maqu			Naqu			Binggou		
		R^2	MBE	RMSE	R^2	MBE	RMSE	R^2	MBE	RMSE
00	Before	0.97	2.90	3.69	0.98	1.46	2.39	0.93	−3.65	4.48
	After	0.97	2.81	3.64	0.98	1.31	2.26	0.93	−2.58	3.55
03	Before	0.98	1.13	1.58	0.93	−0.68	2.35	0.94	−2.33	3.42
	After	0.99	1.01	1.47	0.94	−0.82	2.38	0.94	−1.43	2.83
06	Before	0.97	0.42	1.11	0.92	−1.0	2.18	0.95	−0.23	3.20
	After	0.98	0.35	1.09	0.93	−1.16	2.22	0.96	0.77	2.29
09	Before	0.98	0.36	1.07	0.96	−0.64	1.48	0.97	0.57	2.15
	After	0.98	0.31	1.02	0.96	−0.80	1.53	0.97	1.52	2.54
12	Before	0.96	2.42	3.06	0.97	0.86	1.71	0.95	−0.53	3.02
	After	0.95	2.39	3.01	0.97	0.72	1.63	0.95	0.48	3.21
15	Before	0.94	3.30	4.04	0.97	1.90	2.62	0.93	−1.47	2.54
	After	0.94	3.27	4.0	0.97	1.75	2.45	0.94	−0.60	2.19
18	Before	0.95	3.18	3.87	0.97	1.85	2.63	0.94	−2.58	3.49
	After	0.95	3.12	3.82	0.98	1.70	2.51	0.94	−1.71	2.70
21	Before	0.96	3.23	3.94	0.98	1.84	2.62	0.92	−3.33	4.31
	After	0.96	3.17	3.89	0.98	1.69	2.49	0.92	−2.20	3.20

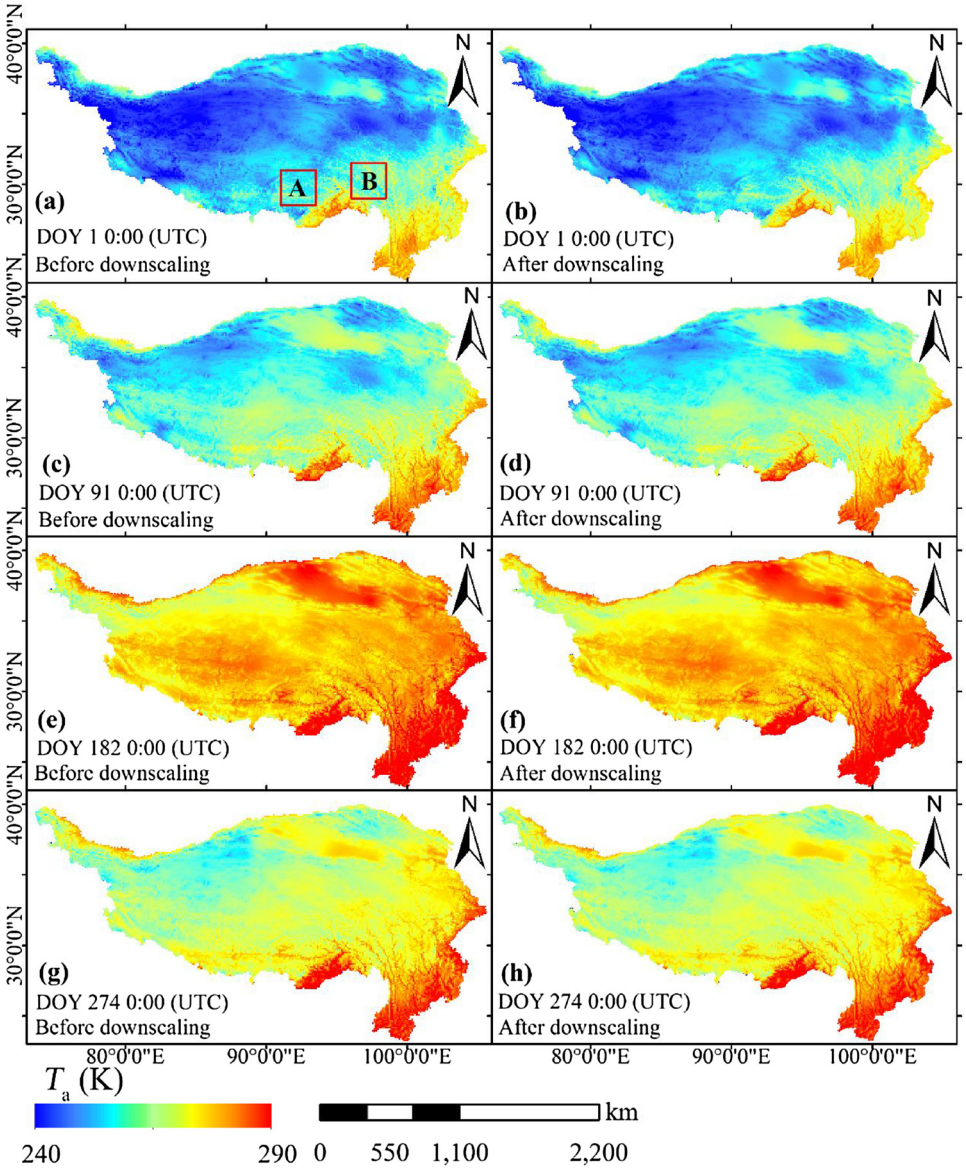


Fig. 10. The CRSMFD T_a before and after downscaling over the entire Tibetan Plateau in 2010.

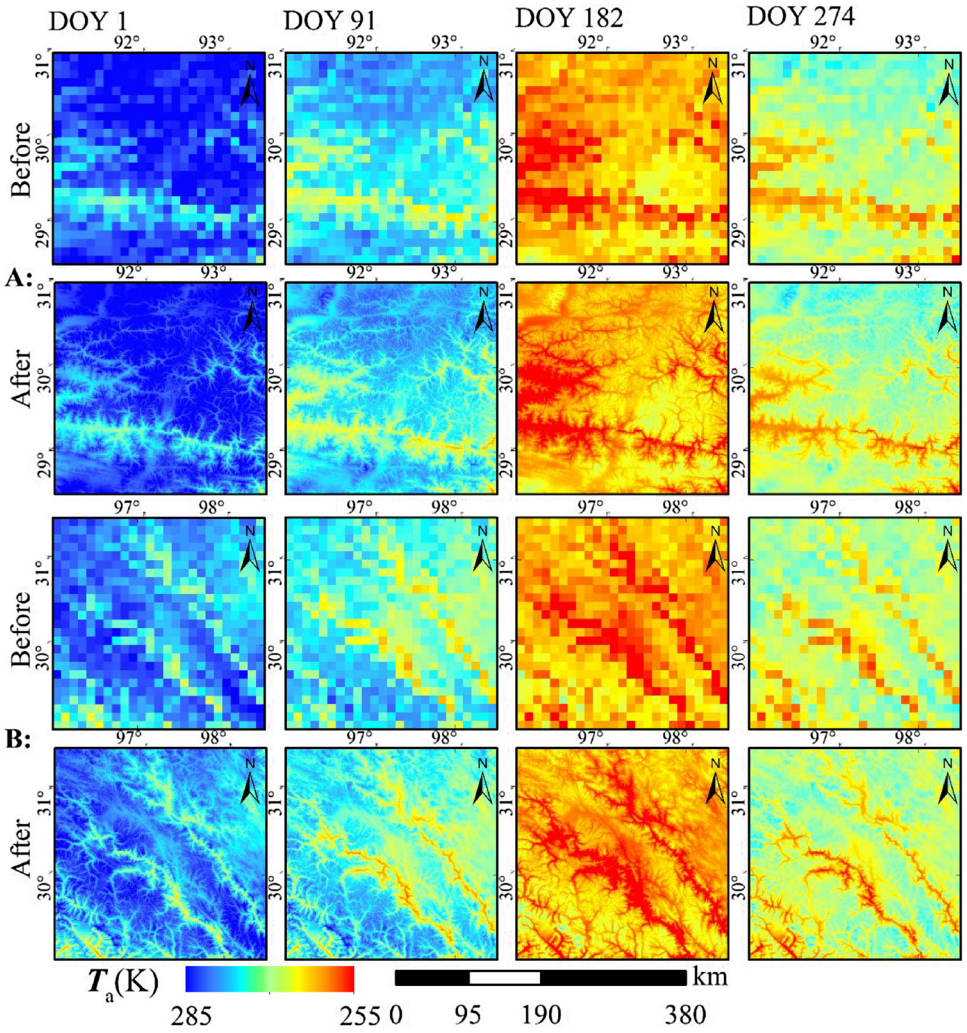


Fig. 11. Before and after downscaling, the CRSMFD instantaneous T_a of local areas.

not only reflect the T_a changes in local areas, but also maintain the overall accuracy and spatial distribution of the original data.

5. Conclusion

T_a is a critical parameter in various studies such as climate change, hydrology, and ecology. However, the coarse-spatial resolutions of current T_a products are unable to meet the ever-increasing demands from related studies and applications. By selecting the Tibetan Plateau as the study area, this study has proposed a practical method to downscale the T_a dataset provided by reanalysis products from coarse resolutions to a medium resolution (i.e. 0.01°). The method is applied to both the daily mean T_a and the 3-hourly instantaneous T_a.

The downscaled 0.01° T_a is firstly evaluated based on the measurements at 105 meteorological stations and three experimental stations. Results show that the daily mean T_a downscaled from the CRSMFD product has a RMSE of 1.13 ± 1.0 K for the 105 stations and RMSEs of 0.96 K to 2.34 K for the three experimental stations; the

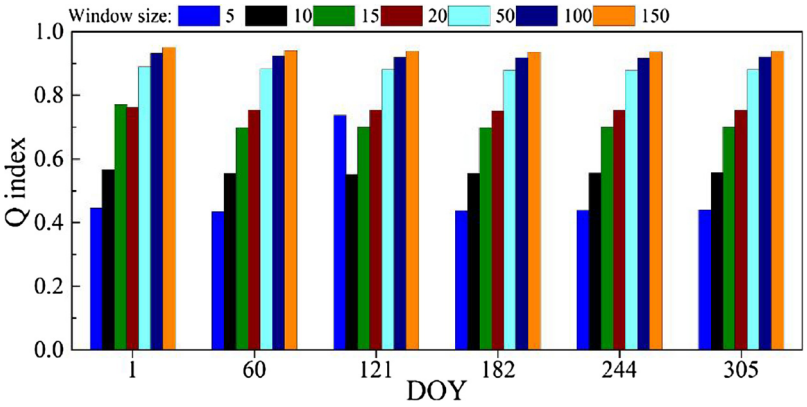


Fig. 12. The Q index calculated with moving windows of different sizes on six days.

instantaneous T_a downscaled from the CRSMFD product has RMSEs of 1.02 K to 4.0 K for the three experimental stations. Compared with T_a before downscaling, T_a after downscaling has better agreement with the ground measured T_a, especially in mountain areas. Furthermore, the downscaled T_a image is evaluated from the perspective of image quality. The evaluation shows that the T_a image after downscaling has good image quality and can clearly reflect the spatial variations of T_a in mountain areas. However, T_a downscaled from the ERAI T_a product has unacceptable accuracy due to the great uncertainty and the coarse original resolution of the ERAI T_a.

Compared with existing methods for downscaling T_a, the mechanism of the proposed method in this paper does not rely on the ground measurements. Therefore, it can be readily applied to large areas. Meanwhile, based on the proposed method, the long-time series of T_a dataset with a 0.01° resolution can be extended from 2000 to 2015 in this study to 1979–2015, which would be beneficial to the long-term analysis of climate change over the Tibetan Plateau.

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